COLLEGE COSTS, STUDENT DEBT, AND STATE-LEVEL EDUCATION PATTERNS IN THE U.S.

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PROJECT OVERVIEW

This project analyzes how average in-state tuition and student debt vary across U.S. states. It examines how these differences relate to household income and educational attainment. The goal is to understand how affordability and outcomes differ across states.

Goals:

- Compare college affordability using tuition, debt, and income data
- Identify relationships between income, education levels, and tuition
- Use machine learning to group states into meaningful patterns

METHODS AND ANALYSIS

- Data cleaning and transformation using pandas
- State-level aggregation of college data
- Creation of affordability ratios such as tuition relative to income
- Correlation analysis between key variables
- Linear regression modeling
- K-means clustering

KEY INSIGHTS

- **Insight 1: Affordability depends on income**

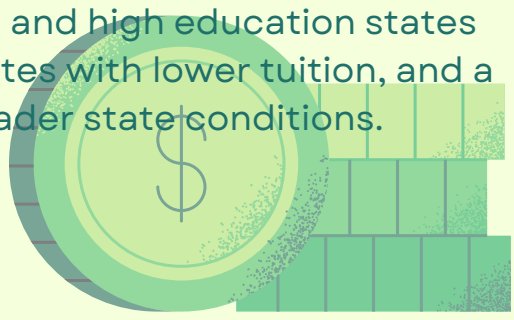
Tuition alone does not determine affordability. States with lower income can still have a higher tuition burden. This shows that cost must be considered relative to income.

- **Insight 2: Education level strongly relates to tuition**

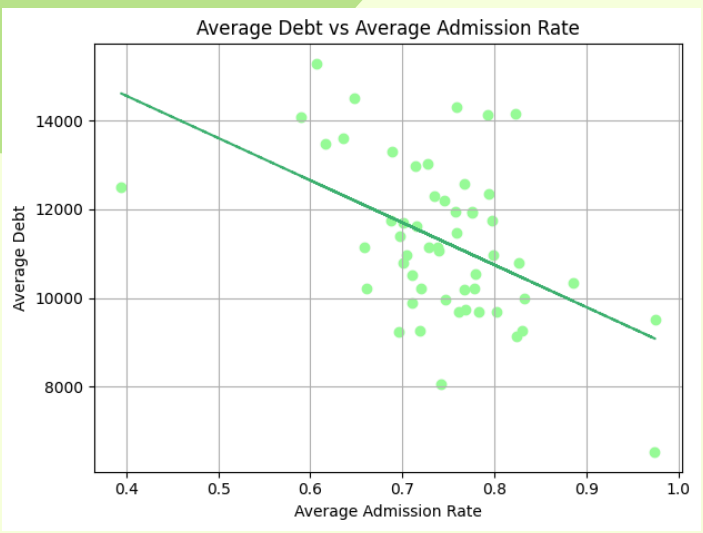
Linear regression shows a strong positive relationship between the percentage of residents with bachelor's degrees and tuition. States with more educated populations tend to have higher tuition, while income alone has a weaker effect.

- **Insight 3: States fall into clear groups**

Clustering grouped states into three tiers: high income and high education states with high tuition, lower income and lower education states with lower tuition, and a middle group. This shows that college costs reflect broader state conditions.

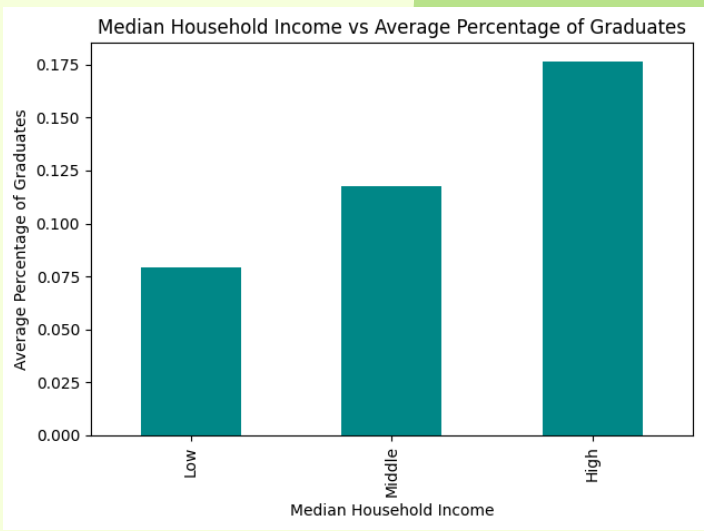


GRAPHS



This scatter plot shows that states with lower admission rates, meaning more selective schools, tend to have higher average student debt. It highlights a negative relationship between selectivity and debt levels.

This bar chart shows that states with higher median household incomes tend to have a higher percentage of residents with advanced degrees. It highlights a strong connection between income and education levels.



CONCLUSION



College costs are not random across the United States. They are closely tied to differences in income levels and educational attainment. States with stronger economies and higher education levels tend to have higher tuition, which affects affordability and student outcomes.

SOURCES

- College Scorecard
- U.S. Census ACS API
- Wikipedia: Educational Attainment by State